Experiment No. 4

Title: Text-to-Image Generation using a Pre-trained Diffusion Model

Aim: To implement a Generative AI image-generation system using a pre-trained diffusion model and generate images based on natural-language prompts provided by the user.

Objectives:

1. To understand the basic concepts of text-to-image generation and Generative AI.
2. To understand the basic working principle of diffusion-based image-generation models.
3. To implement a text-to-image generation system using Hugging Face Diffusers and a pre-trained Stable Diffusion XL model.
4. To create an interactive image-generation interface using Gradio and analyze the effect of text prompts and generation parameters.

Pre-Requisites:

1. Basic knowledge of Python programming.
2. Basic understanding of Deep Learning and neural networks.
3. Basic understanding of Generative AI concepts.
4. Familiarity with Python libraries and Google Colab.
5. Basic knowledge of GPU-based execution in Google Colab is desirable.

Theory:

Text-to-Image Generation: Text-to-image generation is a Generative AI task in which a model creates a new image based on a natural-language description. The user provides a text prompt describing the desired image, and the model generates an image corresponding to that description.

Diffusion Models: Diffusion models are Generative AI models that learn to generate data through a denoising process. During generation, the model starts from a noisy representation and progressively removes noise to produce a meaningful image. When text conditioning is used, the denoising process is guided by the user's text prompt.

Pre-trained Diffusion Models: Training a high-quality image-generation model from scratch requires very large datasets and substantial computational resources. A pre-trained diffusion model can instead be loaded and used directly to generate images from prompts.

Stable Diffusion XL: Stable Diffusion XL (SDXL) is a text-to-image diffusion model that can generate images from natural-language prompts. This experiment uses the pre-trained stabilityai/stable-diffusion-xl-base-1.0 model through Hugging Face Diffusers.

Prompt Engineering: A prompt is the natural-language description supplied to the image-generation model. A useful prompt can specify the subject, environment, style, lighting, composition and desired visual characteristics.

Gradio Interface: Gradio is used to create a simple interactive interface around the Python image-generation function. The user enters a prompt, selects generation parameters, clicks Generate Image, and views the generated image.

Tools and Technologies Required

1. Google Colab
2. Python 3
3. PyTorch
4. Hugging Face Transformers
5. Hugging Face FLAN-T5 pre-trained model
6. Accelerate and SentencePiece
7. Gradio
8. GPU support (optional, but recommended for faster execution)

Procedure:

1. Open Google Colab and create a new notebook.
2. Select a GPU runtime from Runtime → Change runtime type.
3. Install Diffusers, Transformers, Accelerate, Safetensors and Gradio.
4. Import PyTorch, Diffusers and Gradio.
5. Check whether a CUDA-enabled GPU is available.
6. Load the pre-trained stabilityai/stable-diffusion-xl-base-1.0 model using DiffusionPipeline.
7. Move the model to the GPU and use float16 for efficient execution.
8. Create a Python function that accepts a text prompt.
9. Pass the prompt to the diffusion pipeline.
10. Generate an image using the specified number of inference steps and guidance scale.
11. Display the generated image.
12. Test the system using different natural-language prompts.
13. Create a Gradio interface containing prompt, inference-step, guidance-scale and image-output components.
14. Run the Gradio application and generate images interactively.
15. Observe how prompts and parameter values affect the generated image.

Conclusion:

The experiment successfully demonstrated text-to-image generation using a pre-trained diffusion model. Students learned how a natural-language prompt can be converted into a new image using a Generative AI model. The experiment also introduced diffusion-based image generation, prompt engineering, important generation parameters, Hugging Face

Diffusers and Gradio-based interaction. This experiment provides a foundation for understanding advanced image-generation systems and modern Generative AI applications.